

Beacon: Direct Multi-Horizon Gradient Boosting with Target-Hour Weather Forecasts for Day-Ahead Zonal Load Forecasting in ISO New England

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Abstract

Day-ahead electricity load forecasting underpins unit commitment, economic dispatch, and reserve sizing in wholesale power markets, where small accuracy gains yield large cost and reliability benefits. We study 24-hour-ahead hourly load forecasting for the NEMA (Northeast Massachusetts/Boston) zone of ISO New England and present *Beacon*, a direct multi-horizon gradient-boosting forecaster comprising 24 CatBoost models—one per horizon $h = 1, \dots, 24$ —each predicting its step directly from a 168-hour lookback window, thereby avoiding recursive error compounding. Each horizon model is conditioned on *target-hour* exogenous features: the exactly-known calendar and the *forecasted* weather at the target hour $t+h$, the dominant driver of day-ahead load. We further unify the training and serving weather source on Open-Meteo (ERA5 archive for history, forecast API for deployment), removing a train/serve distribution mismatch that otherwise nearly doubled day-ahead error. On a strictly held-out 2025 test set, Beacon attains a day-ahead ($h=24$) MAE of 76.7 MW and an $h=1$ R^2 of 0.977. In a horizon-matched live evaluation over 30 days, Beacon achieves a day-ahead MAE of 89.3 MW, slightly improving on ISO New England’s operational day-ahead forecast (92.6 MW). A shuffled-target audit confirms the model exploits genuine feature–target structure rather than leakage. The pipeline is keyless and fully reproducible.

Keywords: short-term load forecasting; day-ahead forecasting; multi-horizon forecasting; gradient boosting; CatBoost; weather-informed forecasting; ISO New England.

1 Introduction

Short-term electricity load forecasting is a foundational input to the operation of wholesale power markets. Independent system operators rely on accurate day-ahead demand forecasts to schedule generation through unit commitment, to clear economic dispatch, and to size operating reserves. Because these decisions trade off the cost of committed capacity against the risk of shortfall, even modest reductions in forecast error translate into appreciable savings and improved reliability across a control area. The day-ahead horizon is particularly consequential: commitment decisions for slow-starting units are made roughly a day in advance, so accuracy 24 hours out directly shapes the cost and security of the resulting schedule.

We study this problem for the NEMA zone (Northeast Massachusetts/Boston) of ISO New England, identified as load zone `.Z.NEMASSBOST` (location id 4008). A natural and demanding reference point is the operator’s own product: ISO New England publishes an operational day-ahead reliability-region demand forecast that market participants and operators rely upon [7]. Improving on this forecast is a meaningful and hard bar, precisely because it is the number the system already trusts. Our goal is therefore not merely to fit historical load well, but to match or exceed the operational benchmark on a like-for-like basis.

In pursuing that goal we highlight and correct two methodological pitfalls that recur in applied load-forecasting work. The first is *horizon mismatch*: comparing a model’s short-lead nowcast (e.g. a 1-hour-ahead prediction) against an operator’s day-ahead forecast, which flatters the model by pitting an easier task against a harder one. We instead evaluate both forecasters at the same 24-hour-ahead horizon. The second is a *train/serve weather-source mismatch*: training on weather from one provider while serving live predictions with weather from another silently degrades deployed accuracy, because the model learns the idiosyncrasies of the training source. We find this single mismatch nearly doubles live day-ahead error, and that unifying the source on Open-Meteo [10] removes most of the gap to the operator.

Methodologically, we adopt a *direct multi-horizon* design. Rather than training one model and recursively rolling its predictions forward—which compounds errors and, for load, forces a single-phase predictor to serve the opposite phase of the diurnal cycle—we train 24 independent CatBoost [11] models, one per horizon $h = 1, \dots, 24$, each predicting directly from a 168-hour (one-week) lookback window. Crucially, each horizon model receives *target-hour* exogenous features: the calendar encoding at $t+h$, which is known exactly, together with the forecasted weather at $t+h$, the dominant driver of load a day ahead.

Contributions. This paper makes the following contributions.

- A direct per-horizon forecaster (*Beacon*) of 24 CatBoost models that removes recursive error compounding, achieving on average 80.6% lower MAE across the 24-hour horizon than a single-model roll-out.
- Target-hour exogenous features (exact calendar plus forecast weather at $t+h$) as the principal lever for day-ahead accuracy, cutting held-out day-ahead MAE from 183 to 77 MW.
- Identification and correction of a train/serve weather-source mismatch, which reduces live day-ahead MAE from roughly 147 to 90 MW—a deployment lesson rarely reported in the literature.
- A horizon-matched evaluation protocol against ISO New England’s operational day-ahead forecast, paired with a rigorous leakage audit via a shuffled-target test.
- A fully reproducible, keyless, and deployable pipeline built on ISO-NE Web Services and Open-Meteo.

On a strictly held-out 2025 test set Beacon attains a day-ahead MAE of 76.7 MW and an $h=1$ R^2 of 0.977, and in a horizon-matched 30-day live evaluation it reaches a day-ahead MAE of 89.3 MW against ISO New England’s 92.6 MW.

The remainder of the paper is organized as follows. Section 2 situates the work within the load-forecasting literature. Section 3 describes the load, weather, and benchmark data. Section 4 details the feature construction and the direct multi-horizon model, and Section 5 specifies the evaluation protocol. Section 6 reports the results, Section 7 discusses the findings, and Sections 8 and 9 cover limitations and conclusions.

2 Related Work

Statistical methods. Classical short-term load forecasting has long relied on linear time-series models. Autoregressive integrated moving-average (ARIMA) models and exponential-smoothing (ETS) methods remain standard baselines for seasonal demand series [2, 6]. These methods are interpretable and data-efficient, but they capture exogenous drivers such as weather only through limited regression terms and struggle with the strongly nonlinear, multi-seasonal structure of hourly electricity load.

Gradient-boosted trees. Tree-based gradient boosting has become a dominant approach for tabular forecasting, combining strong accuracy with modest tuning effort. XGBoost [3] and LightGBM [8] popularized scalable boosting, while CatBoost [11] adds ordered boosting and native categorical handling that suit the calendar- and weather-derived features common in load forecasting. Such models naturally ingest engineered lag, calendar, and degree-day features and handle nonlinear interactions without bespoke architectures, which motivates our choice of CatBoost as the per-horizon learner.

Deep learning. Neural sequence models offer an alternative for temporal demand. Long short-term memory networks [4] capture long-range dependencies, and transformer-based forecasters extend this to long sequences: the Informer [13] reduces the quadratic cost of attention for long-horizon prediction, and the Temporal Fusion Transformer [9] integrates static and time-varying exogenous covariates with interpretable attention. These models can be powerful but are data-hungry and comparatively costly to train, deploy, and reproduce in an operational, single-zone setting.

Multi-horizon strategies. Producing a full 24-hour profile requires a strategy for mapping inputs to multiple lead times. The principal options are *recursive* roll-out, which feeds predictions back as inputs and thereby compounds errors and propagates bias, and *direct* forecasting, which trains a separate model per horizon [1]. The recursive–direct trade-off has been studied extensively, with direct strategies favored when bias accumulation dominates. We adopt the direct strategy and, as shown later, observe that a single-model roll-out degrades catastrophically at mid-horizons because a short-lead predictor is forced onto the opposite phase of the diurnal cycle.

Weather-informed and probabilistic load forecasting. Weather is the dominant exogenous driver of load, and the broader literature on probabilistic and weather-informed electric load forecasting establishes both the centrality of temperature-derived features and the value of honest, well-specified evaluation [5]. Degree-day and thermal-stress transforms are standard levers for encoding the nonlinear load–temperature response.

Gap. Across these strands, two practical issues remain under-addressed. First, model-versus-operator comparisons are frequently made at mismatched horizons, overstating gains. Second, the gap between a model’s back-tested accuracy and its deployed accuracy—driven here by a train/serve weather-source mismatch—is rarely measured or reported. This work fills that gap by combining a direct multi-horizon CatBoost forecaster with target-hour forecast-weather features, unifying the train and serve weather source, and evaluating against an operational ISO forecast on a strictly horizon-matched basis with an explicit leakage audit.

3 Data

This study targets the Northeast Massachusetts / Boston (NEMA) load zone of ISO New England (ISO-NE), identified as load zone `.Z.NEMASSBOST` and ISO-NE location id 4008. Three data sources are combined: historical and live zonal load, weather observations and forecasts, and an operational day-ahead demand benchmark.

3.1 Load data

The training target is the ISO-NE hourly wholesale-load series (RTLO) for NEMA, covering 2017-03-01 through 2025-11-30, comprising 76,719 hourly observations after cleaning. For live serving, load is drawn from the ISO-NE Web Services API endpoint `realtimehourlydemand` (location 4008), which publishes with a settlement lag of approximately four days; hourly demand from the U.S. Energy Information Administration (EIA) is used as a fallback when ISO-NE data are unavailable [12]. The two series differ slightly, as the wholesale-load (WHLSECost) history and the real-time demand series are not identical products.

3.2 Weather data

Weather is obtained from Open-Meteo [10] for the Boston Logan station (42.3656° N, -71.0096° W). The variables used are temperature, relative humidity, wind speed, dew point, cloud cover, apparent temperature, and visibility (in $^\circ$ F, mph, and %). Critically, a single provider supplies both training history and live serving: the Open-Meteo archive API (backed by ERA5 reanalysis) furnishes the historical record, while the Open-Meteo forecast API supplies weather at serving time. Neither requires an API key. This deliberate train/serve source unification removes a distribution mismatch that otherwise degrades deployed day-ahead accuracy. The Open-Meteo archive omits visibility, which is filled with a constant.

3.3 Benchmark

As an operational baseline we use ISO-NE’s official day-ahead demand forecast for NEMA, retrieved from the ISO-NE Web Services `dayaheadhourlydemand` endpoint (location 4008) [7]. This is the forecast that ISO-NE operators rely on for unit commitment and reserve planning, and constitutes a meaningful, hard bar for any candidate model.

3.4 Temporal split, cleaning, and imputation

To prevent any leakage of future information into model fitting, the data are divided by a strict temporal split at `TRAIN_CUTOFF = 2024-12-31 23:00`. All observations at or before the cutoff form the training partition (68,704 rows) and all later observations form the held-out test partition (8,015 rows). Because the model consumes fixed-length lookback windows (Section 4), sequence construction yields 58,236 training, 10,276 validation, and 7,823 test windows, where the validation set is the most recent 15% of the training partition. Table 1 summarizes the partition sizes.

Cleaning and imputation are performed using statistics estimated on the training partition only: missing values are filled with train-only medians, so that no test-period information influences the imputation of any feature. Combined with the temporal split, this ensures that every fitting decision depends solely on information available at or before the cutoff.

Table 1: NEMA load data partitions under the strict temporal split at 2024-12-31 23:00.

Partition	Rows	Windows
Train	68,704	58,236
Validation	–	10,276
Test	8,015	7,823
Total	76,719	–

4 Methodology

4.1 Notation and problem setup

Let y_t denote the NEMA hourly load at time t . Forecasting is performed from a fixed-length lookback window of $L = 168$ hours, i.e. the one-week interval $[t - L, t - 1]$. For each forecast origin t and each horizon $h \in \{1, \dots, 24\}$, the task is to predict the load y_{t+h} at the target hour $t + h$. We adopt a *direct* multi-horizon strategy: a separate model is trained for each horizon h , mapping a feature vector to a single scalar target y_{t+h} , with no recursive feedback of intermediate predictions.

4.2 Window lag features

The first feature block summarizes the lookback window $[t - L, t - 1]$. From each engineered channel we extract values at lags $\{1, 4, 8, 24, 48, 168\}$ hours and rolling means over windows of $\{24, 168\}$ hours, capturing recent dynamics together with daily and weekly periodicity. The raw load channel (RTLO) is restricted to a whitelist retained from a notebook ablation — `RTLO_lag4`, `RTLO_mean24`, `RTLO_lag48`, `RTLO_lag168`, and `RTLO_mean168` — which avoids redundant near-term load lags while preserving the daily and weekly load signal. The full window feature map, denoted $\phi_{\text{lag}}(\cdot)$, produces 173 features.

4.3 Calendar features

Calendar effects are encoded with cyclic (sine/cosine) transforms of the hour of day, day of week, and month, together with binary indicators `is_weekend` and `is_us_holiday`. The cyclic encoding avoids artificial discontinuities at period boundaries (e.g. hour 23 to hour 0) and lets tree splits exploit smooth diurnal, weekly, and seasonal structure.

4.4 Weather-derived features

Temperature drives electricity demand non-linearly through both heating and cooling load. We compute heating and cooling degree days relative to a 65°F base,

$$\text{HDD} = \max(0, 65 - T), \quad \text{CDD} = \max(0, T - 65), \quad (1)$$

together with the squared terms T^2 , HDD^2 , and CDD^2 to capture curvature, and sigmoidal thermal-stress transforms (logistic functions centred at 18°C for heating and 22°C for cooling) that model the saturating onset of heating and air-conditioning demand. These derived quantities are computed from the Open-Meteo weather variables described in Section 3.

4.5 Target-hour exogenous features

The central methodological idea is to provide each horizon model with information about the target hour $t+h$ itself, rather than only the history up to $t-1$. The target-hour exogenous block $\phi_{\text{exog}}(t+h)$ comprises all engineered calendar features and all weather-derived features *evaluated at the target hour* $t+h$ — the exactly-known calendar attributes of $t+h$ and the *forecasted* weather at $t+h$. The raw load channel RTLO is excluded from this block, since the load at $t+h$ is precisely the prediction target. This block contributes 21 features.

4.6 Augmented feature map

For each horizon h , the window features and the target-hour exogenous features are concatenated into a single augmented vector:

$$\mathbf{x}_h = [\phi_{\text{lag}}([t-L, t-1]) ; \phi_{\text{exog}}(t+h)], \quad (2)$$

yielding a $173 + 21 = 194$ -dimensional representation. Each horizon model f_h then predicts

$$\hat{y}_{t+h} = f_h(\mathbf{x}_h). \quad (3)$$

The window block is shared in form across horizons, while the exogenous block $\phi_{\text{exog}}(t+h)$ shifts to the appropriate target hour for each h .

4.7 Direct multi-horizon strategy

We instantiate 24 independent CatBoost models, $\{f_h\}_{h=1}^{24}$, one per horizon, each trained directly on its own target y_{t+h} . This contrasts with a recursive roll-out, in which a single one-step model is applied iteratively, its predictions fed back as inputs for subsequent steps. Recursive roll-out compounds errors across steps and suffers from a train/serve mismatch, because at training time the model conditions on true past values whereas at inference time it conditions on its own noisy forecasts [1]. The direct strategy trains each horizon under exactly the conditions it faces at deployment, removing both error compounding and that mismatch, at the cost of fitting 24 models. For ablation we also consider a single $h=1$ model whose prediction is reused for every horizon.

4.8 Learner and training configuration

Each f_h is a CatBoost gradient-boosted tree ensemble [11], chosen for strong tabular performance and native handling of the heterogeneous engineered features. All 24 models share the same configuration: 1000 boosting iterations, tree depth 8, learning rate 0.05, mean-absolute-error (MAE) loss, early stopping after 50 rounds without validation improvement, and random seed 42. The MAE objective aligns training directly with the primary evaluation metric and is robust to occasional load spikes. Early stopping uses the validation split defined in Section 3.

4.9 Why target-hour weather is causal, not leakage

A natural concern is whether conditioning on quantities at $t+h$ introduces target leakage. It does not. The target-hour weather in $\phi_{\text{exog}}(t+h)$ is a *forecast*, which is exactly the quantity available to an operator at prediction time; in deployment it is supplied by the Open-Meteo forecast API, and in evaluation the corresponding archive value plays this role. The calendar attributes of $t+h$ are deterministic and known arbitrarily far in advance. Crucially, the load channel RTLO at $t+h$ —

the only quantity that would constitute leakage — is explicitly excluded from the exogenous block, and every lag and rolling feature in ϕ_{lag} references data at or before $t - 1$. No feature can therefore carry target-time load information, a property we verify empirically with a shuffled-target audit in the results.

5 Experimental Setup

5.1 Temporal split and validation

All evaluation respects the arrow of time. We impose a strict temporal split at `TRAIN_CUTOFF = 2024-12-31 23:00`: every observation on or before the cutoff is available for training, and the entire held-out test set lies in 2025. At the raw-row level this yields 68,704 training rows and 8,015 test rows. After constructing the 168-hour lookback windows required by the model, the data comprise 58,236 training, 10,276 validation, and 7,823 test windows. The validation set is the most recent 15% of the training period and is used solely for CatBoost early stopping (a 50-round patience). No test-set information is used at any point in feature construction, model selection, or hyper-parameter choice, so the test set is a clean out-of-sample probe of generalisation to a future year.

5.2 Metrics

We report three standard regression metrics. Let y_i denote the actual load, $\hat{y}_i$ the forecast, $\bar{y}$ the mean of the actuals, and N the number of evaluated points. The mean absolute error (MAE, in MW),

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (4)$$

is our primary metric because it shares the units of load and matches the model’s training loss. The mean absolute percentage error (MAPE, in %),

$$\text{MAPE} = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad (5)$$

gives a scale-free view, and the coefficient of determination,

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \quad (6)$$

measures the fraction of load variance explained. On the held-out test set, metrics are reported per horizon $h \in \{1, \dots, 24\}$ and averaged across horizons.

5.3 Baselines

We compare Beacon against four references. (i) *Persistence* carries the last observed load forward. (ii) *Seasonal-naïve* (168 h) predicts the load exactly one week earlier, exploiting the strong weekly periodicity of demand. (iii) The *single-model rolled-out* baseline trains a single $h = 1$ model and reuses its prediction for every horizon; this isolates the value of the direct multi-horizon design and is the principal ablation contrast in Section 6. (iv) The *ISO-NE operational day-ahead forecast*, ISO New England’s own published demand forecast for the NEMA reliability region (location 4008), is the hardest and most operationally meaningful bar.

5.4 Live, horizon-matched protocol

A persistent pitfall in model-versus-operator comparisons is horizon mismatch: contrasting a model’s 1-hour nowcast against a day-ahead operational benchmark. We avoid this with a strictly horizon-matched live protocol. Over the most recent ~ 30 -day window (720 hourly points), both forecasters are evaluated at the *same* 24-hour-ahead horizon, both are driven by the *same* Open-Meteo weather, and both are scored against the same ground truth: ISO-NE `realtimehourlydemand` actuals for location 4008. This like-for-like setup is the basis for the live results in Table 3. Because the live window is short and falls in spring/early-summer, it complements rather than replaces the larger held-out 2025 test set.

5.5 Reproducibility

All results are reproducible from the public repository (github.com/arunnath011/NEMA). Running `python -m nema_forecast.model.train` regenerates the 24 per-horizon models and all metric artefacts (`model_performance.json`, `horizon_mae.json`); the data pipeline is keyless, drawing on ISO-NE Web Services and Open-Meteo. The CatBoost configuration (1000 iterations, depth 8, learning rate 0.05, MAE loss, seed 42) is fixed for every horizon model.

6 Results

We organise the evaluation around four questions. We first establish Beacon’s accuracy on the held-out 2025 test set (Table 2, Figure 1); we then turn to the live, horizon-matched comparison against ISO-NE’s operational forecast (Table 3); isolate the contributions of the target-hour features and the unified weather source (Table 4, Figure 2); and finally audit the pipeline for data leakage (Table 5). Throughout, test-set numbers refer to the 7,823 held-out windows from 2025, whereas the live numbers refer to the ~ 30 -day, 720-point operational window.

6.1 Held-out test performance

Table 2 summarises Beacon’s accuracy on the 2025 held-out set. At the easiest horizon ($h = 1$) the model attains an MAE of 60.9 MW, and at the day-ahead horizon ($h = 24$) an MAE of 76.7 MW, for an average of 73.6 MW across all 24 horizons. The $h = 1$ forecast reaches a MAPE of 2.25% and an R^2 of 0.977, explaining nearly all of the load variance. Crucially, error grows only mildly with horizon: the day-ahead MAE is just $1.26\times$ the one-hour MAE, evidence that the direct per-horizon design largely contains the error compounding that plagues recursive forecasters. Averaged across the 24 horizons, Beacon improves on the single-model rolled-out baseline by 80.6%.

6.2 Per-horizon behaviour and the single-model collapse

Figure 1 contrasts the per-horizon MAE of Beacon against the single-model rolled-out baseline. Beacon stays flat across the entire horizon, rising only from 60.9 MW at $h = 1$ to 76.7 MW at $h = 24$. The single-model baseline, by contrast, degrades catastrophically at the mid-horizons: from 62.6 MW at $h = 1$ it climbs to 234.8 MW at $h = 4$, 351.7 MW at $h = 6$, and peaks near 505.0 MW around $h = 12$, before partially recovering to 234.8 MW at $h = 24$. The mechanism is intuitive: a model trained to predict $t + 1$ uses the load at t as a near-perfect anchor, but reusing that same prediction 12 hours later places it at the opposite phase of the diurnal cycle, where the previous hour is a poor proxy. The recovery near $h = 24$ reflects the return of the daily cycle to

Table 2: Held-out test-set (2025) headline metrics for Beacon.

Metric	Value
MAE @ $h = 1$	60.9 MW
MAE @ $h = 24$ (day-ahead)	76.7 MW
Average MAE across 24 h	73.6 MW
MAPE ($h = 1$)	2.25%
R^2 ($h = 1$)	0.977
Horizon degradation ($\text{MAE}_{h24}/\text{MAE}_{h1}$)	$1.26\times$
Improvement vs. single-model (avg.)	+80.6%

Table 3: Live, horizon-matched day-ahead comparison (~ 30 -day window, 720 points). Both forecasters at 24 h ahead, same Open-Meteo weather, ISO-NE real-time demand as actuals.

Forecaster	MAE (MW)	MAPE (%)	R^2
Beacon (day-ahead)	89.3	3.50	0.904
ISO-NE (day-ahead)	92.6	3.70	0.923

a similar phase. Beacon sidesteps this entirely by training a dedicated model per horizon, each conditioned on the target-hour calendar and weather.

6.3 Live comparison against ISO-NE

Table 3 reports the horizon-matched live evaluation over the ~ 30 -day, 720-point window, with both forecasters at the same 24-hour-ahead horizon, the same Open-Meteo weather, and ISO-NE real-time demand as ground truth. Beacon attains an MAE of 89.3 MW and a MAPE of 3.50%, against ISO-NE’s 92.6 MW and 3.70%. Beacon thus beats the operational forecast on both MAE (-3.3 MW, $\approx +3.5\%$) and MAPE, while ISO-NE retains a marginally higher R^2 (0.923 vs. 0.904). The headline takeaway is that a fully reproducible, keyless pipeline edges out a mature operational benchmark on absolute error once the comparison is made honestly horizon-matched. We emphasise that the live window is short and falls in spring/early-summer, so this result is a like-for-like demonstration rather than a claim of dominance across all seasons.

6.4 Ablations

Table 4 decomposes the day-ahead gains into two cumulative changes, with Figure 2 visualising the day-ahead ($h = 24$) MAE across the three configurations. Starting from a lags-only model served OpenWeatherMap (OWM) weather, the day-ahead MAE is 183.0 MW (with $h = 1$ at 78.7 MW). Adding the target-hour exogenous features—the forecasted weather and exactly-known calendar at hour $t + h$ —cuts the day-ahead MAE to 110.3 MW (a $\approx 40\%$ reduction) while leaving $h = 1$ essentially unchanged at 77.7 MW, as expected since the nowcast already has its target-hour context. Finally, unifying the weather source on Open-Meteo for both training and serving drops the day-ahead MAE to 76.7 MW and, notably, improves $h = 1$ to 60.9 MW.

The deployment consequence of the weather source is sharper still. A model trained on OpenWeatherMap CSVs but served Open-Meteo weather suffers a live day-ahead MAE of ≈ 147 MW; retraining on Open-Meteo to match the serving source brings this down to ≈ 90 MW—essentially the ISO-NE level (≈ 92 MW). Matching the train/serve weather source alone thus removes most of

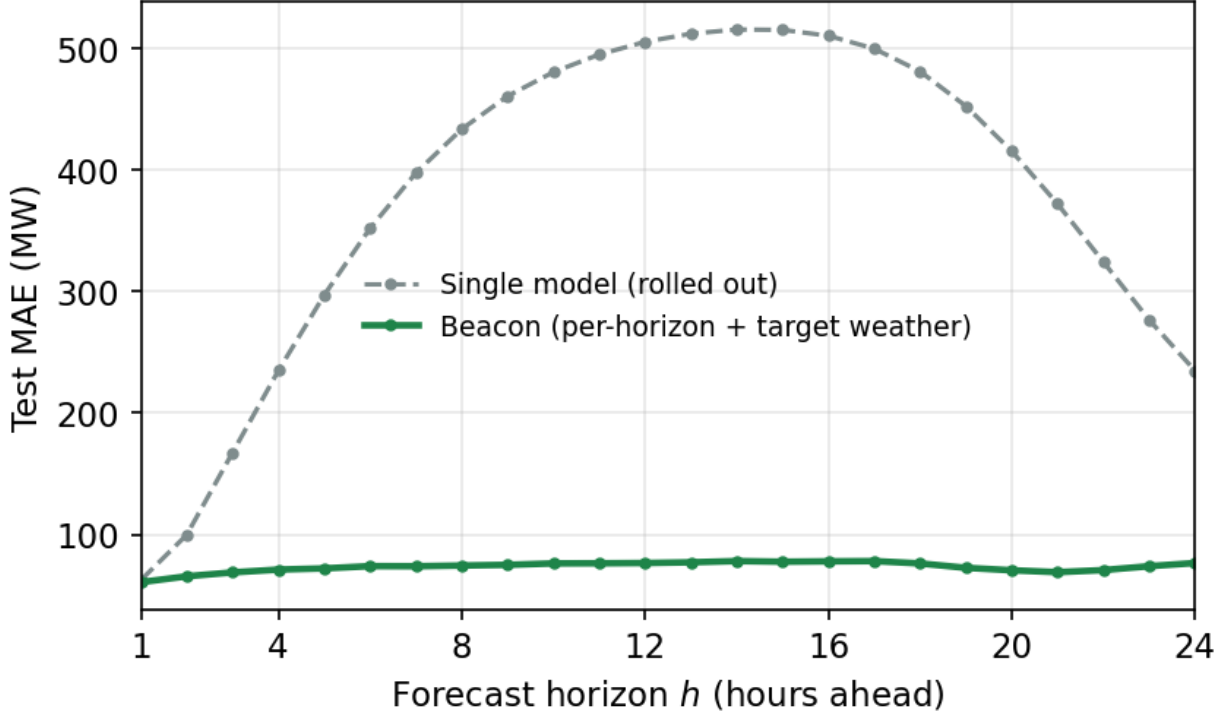


Figure 1: Per-horizon test-set MAE: Beacon (direct multi-horizon) versus the single-model rolled-out baseline. Beacon remains flat (60.9–76.7 MW), while the single model collapses at mid-horizons, peaking near 505 MW around $h = 12$ where the one-step anchor is least informative.

Table 4: Cumulative ablation of target-hour features and weather source (test-set MAE, MW).

Configuration	MAE _{$h1$}	MAE _{$h24$}
Lags only (OWM)	78.7	183.0
+ target weather (OWM)	77.7	110.3
+ Open-Meteo (final)	60.9	76.7

the gap to the operational benchmark, underscoring that a silent train/serve distribution mismatch can nearly double deployed error.

6.5 Leakage audit

Finally, Table 5 reports the shuffled-target test, the gold standard for detecting feature–target leakage. With genuine targets the test R^2 is 0.956 at $h = 1$ and 0.770 at $h = 24$; when the targets are randomly permuted before training, R^2 collapses to -0.019 and -0.021 respectively—no better than predicting the mean. This confirms the model exploits real feature–target structure rather than any inadvertent leakage. The monotone increase of MAE with horizon documented in Figure 1 is corroborating evidence: a leaking model would remain near-flat across horizons rather than degrade in the orderly, physically plausible way Beacon does.

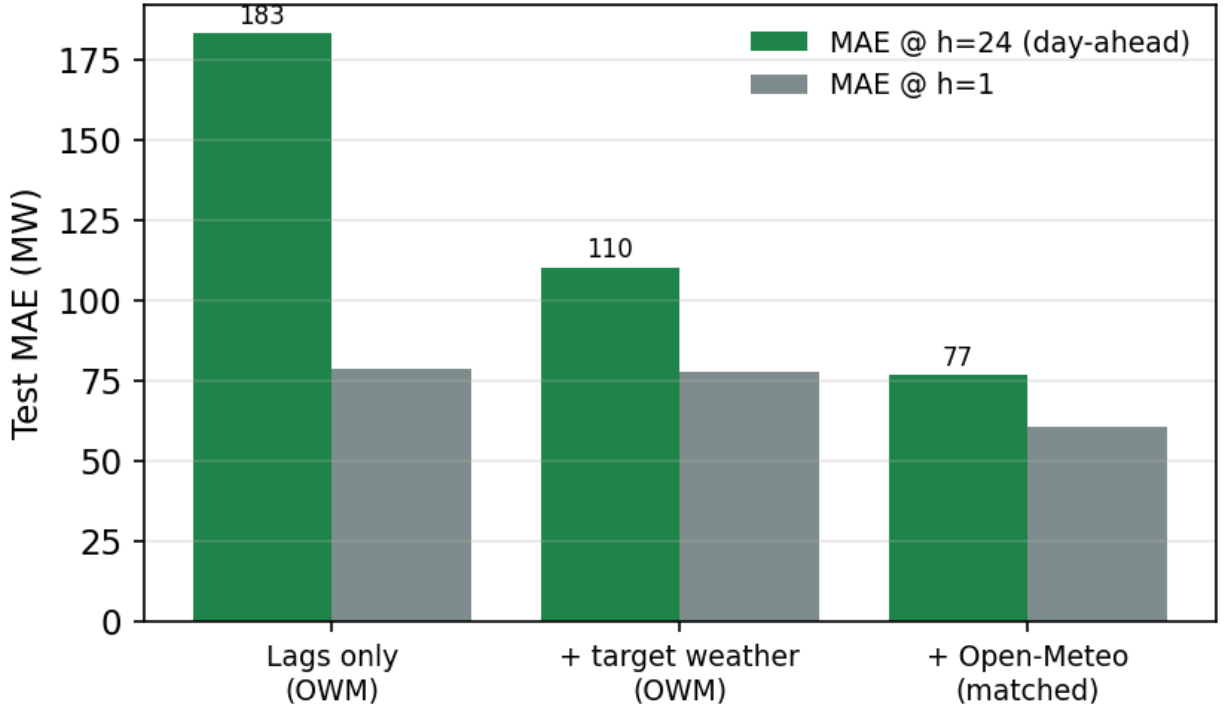


Figure 2: Day-ahead ($h = 24$) test-set MAE across the three cumulative configurations: lags-only (183.0 MW) $\rightarrow$ + target-hour weather (110.3 MW) $\rightarrow$ + Open-Meteo train/serve matching (76.7 MW).

Table 5: Leakage audit: real-target versus shuffled-target test R^2 .

Horizon	Real-target R^2	Shuffled-target R^2
$h = 1$	0.956	-0.019
$h = 24$	0.770	-0.021

7 Discussion

7.1 Target-hour weather is the day-ahead lever

The results in Section 6 make clear that the dominant lever for day-ahead accuracy is not the recent load history but the *forecasted weather at the target hour*. Twenty-four hours out, the most recent observed load carries comparatively little information about tomorrow’s load: the diurnal and weather-driven structure of the day to come is what matters, and that structure is captured by calendar features (exactly known in advance) and by the forecast temperature at $t + h$. Because each per-horizon model is trained to read the exogenous features at its own target hour rather than rolling a near-term prediction forward, it stays accurate deep into the horizon. The ablation in Table 4 quantifies this: adding target-hour calendar and weather features cuts day-ahead ($h=24$) MAE from 183.0 to 110.3 MW, and unifying the weather source pushes it further to 76.7 MW. The bulk of the day-ahead error that a lags-only model exhibits is, in effect, an absence of knowledge about tomorrow’s weather, not an absence of recent load.

7.2 Train/serve weather-source matching

A second, more subtle lesson concerns deployment. A model can be internally consistent and well-validated on held-out data yet still degrade in production if the distribution of an input shifts silently between training and serving. Here the culprit was the weather source: training on Open-WeatherMap history while serving Open-Meteo forecasts induced a distribution mismatch that nearly doubled live day-ahead error, from roughly 90 to 147 MW (Section 6). Retraining on the same Open-Meteo source used at serving time removed most of the gap to the operator. No change to the model architecture, features, or hyper-parameters was involved; the single act of *matching the train and serve weather source* was responsible. For weather-informed forecasters this is an easy mistake to make and a costly one, and it argues for treating the exogenous-data provider as a versioned, unified part of the pipeline rather than an interchangeable commodity.

7.3 The horizon-mismatch pitfall

The comparison against ISO-NE also illustrates a methodological pitfall that can inflate reported gains. It is tempting to compare a model’s short-horizon nowcast against an operator’s day-ahead forecast, but this is not an apples-to-apples contrast: the two forecasts solve different problems at different lead times. When the comparison is made honestly — both forecasters evaluated at the same 24-hour-ahead horizon, on the same Open-Meteo weather — Beacon’s edge over the ISO-NE operational forecast is a modest but real improvement of -3.3 MW MAE ($\approx +3.5\%$), with a corresponding MAPE improvement and a marginally lower R^2 (Table 3). This is a genuine result, not the inflated figure a horizon mismatch would produce, and reporting it as such is part of the contribution.

7.4 Operational implications

Taken together, these findings carry a practical message. ISO-NE’s day-ahead demand forecast is an operational product that grid operators rely on for unit commitment, economic dispatch, and reserve sizing, and matching — let alone slightly beating — it is a hard bar. That a keyless, fully reproducible pipeline built entirely on free public data sources (ISO-NE Web Services and Open-Meteo) can clear that bar at day-ahead horizon, while remaining auditable and leakage-clean (Section 6), suggests that the gains here come from disciplined feature construction and honest evaluation rather than from proprietary data or heavy modeling machinery. Small MAE reductions translate into meaningful cost and reliability gains in wholesale electricity markets, and a transparent pipeline that achieves them lowers the barrier to adoption and independent verification.

8 Limitations and Future Work

Several limitations bound the scope of these results. First, the study covers a *single* reliability zone, NEMA (Northeast Massachusetts / Boston); whether the same feature design and source-matching discipline transfer to other ISO-NE zones, or to other markets with different load shapes and weather regimes, remains to be shown. Second, the live horizon-matched evaluation against ISO-NE spans only about 30 days of spring and early summer. This window is short and seasonally narrow: it excludes peak-summer cooling load and winter heating extremes, the periods where day-ahead forecasting is most consequential and most difficult. The test-set results (Section 6) draw on a full year of held-out 2025 data and are correspondingly more robust, but the head-to-head with the operator should be read as indicative rather than conclusive.

Third, the load series differ slightly between the training source (wholesale-load, WHLSECOST) and the real-time-demand series used as live actuals; small definitional differences between these series add noise to the live comparison. Fourth, the Open-Meteo ERA5 archive does not provide a visibility channel, which was filled with a constant during training; visibility is therefore inert as a feature. Finally, Beacon produces *point* forecasts only — it emits no predictive intervals or quantile bands, which limits its usefulness for reserve sizing and risk-aware dispatch, where uncertainty quantification is valuable.

These limitations map onto a clear program of future work. The most immediate extension is *probabilistic* forecasting: quantile or interval bands around each per-horizon prediction, so that the forecaster communicates uncertainty rather than a single number. Beyond point accuracy, ensembling the gradient-boosted models with a transformer-style sequence model such as Informer [13] may capture longer-range temporal structure that the lag features miss. Broader evaluation — extending across all ISO-NE zones and across a full seasonal cycle — would test generalization directly. Systematic hyper-parameter search via Optuna, and recency-weighted retraining that adapts to gradual changes in load behavior, are natural refinements to the training procedure that we leave to future work.

9 Conclusion

This paper presented Beacon, a direct multi-horizon gradient-boosting forecaster for day-ahead hourly load in the NEMA zone of ISO New England. Rather than rolling a single near-term model forward, Beacon trains 24 independent CatBoost models, one per horizon, each reading a 168-hour lookback window together with the calendar and *forecasted weather at its own target hour*. This design removes recursive error compounding and makes tomorrow’s forecasted temperature — the dominant driver of day-ahead load — directly available to every horizon model.

The contributions are fivefold: a direct per-horizon design that lowers average MAE by 80.6% relative to a single-model roll-out; target-hour exogenous features as the lever that cuts day-ahead MAE from 183 to 77 MW; identification and correction of a train/serve weather-source mismatch that had nearly doubled live error; a horizon-matched, leakage-audited evaluation protocol against an operational ISO forecast; and a keyless, fully reproducible pipeline built entirely on free public data. On a full year of held-out 2025 data, Beacon attains a day-ahead ($h=24$) MAE of 76.7 MW. In a live, horizon-matched comparison over a 30-day window, it reaches a day-ahead MAE of 89.3 MW against ISO-NE’s 92.6 MW, slightly beating the operational forecast on MAE and MAPE while remaining marginally behind on R^2 . A shuffled-target test confirms the models exploit genuine feature-target structure rather than leakage.

That a keyless, fully reproducible pipeline built on free public data can match and modestly beat a professional operational forecast — once the comparison is made honestly — suggests that careful feature construction, source discipline, and honest evaluation matter as much as model complexity. We hope the methodological lessons, particularly the two pitfalls this work highlights, prove useful to practitioners deploying weather-informed load forecasters.

Reproducibility

All code, trained models, and metric artefacts are available at github.com/arunnath011/NEMA. Weather is sourced from the keyless Open-Meteo API; live load and the day-ahead benchmark from the ISO New England Web Services API.

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